Contents

1 Introduction 1
1.3 Notational Conventions 1
1.4 Problem-Solving Approach 1
1.5 Important Concepts from Circuit Theory 1
1.6 Frequency Spectrum of Electronic Signals 1
1.7 Amplifiers 1
1.K Key Takeaways & Watch Outs 1
1.F Chapter 1 Formula Sheet 1
1.C Constants & Key Numbers 1
2 Solid-State Electronics 1
2.1 Solid-State Electronic Materials 1
2.2 Covalent Bond Model 2
2.3 Drift Currents and Mobility 2
2.4 Resistivity of Intrinsic Silicon 2
2.5 Impurities in Semiconductors 2
2.6 Electron and Hole Concentrations in Doped Semiconductors 2
2.7 Mobility and Resistivity in Doped Semiconductors 2
2.8 Diffusion Currents 2
2.9 Total Current 2
2.10 Energy Band Model 2

1 Introduction

1.3 Notational Conventions

[J p.12]

Form	Meaning	e.g.
v_{BE}	total = dc + signal	$V_{BE} = V_{BE} + v_{be}$
V_{BE}	dc (bias / Q-point) value	$V_{BE} = 0.7$ V
v_{be}	signal (ac) part only	$v_{be} = 5\sin\omega t$ mV
V_{be}	amplitude of sinusoid; **bold** = phasor	$\mathbf{V_{be}} = V_{be}\angle\theta$

Rule: CAPITAL subscript ⇒ includes dc; lowercase letter ⇒ varies with time.
Split a signal: dc part = constant term, signal part = everything else.
e.g. $v_A = 5\sin 2000\pi t + 4 + 3\cos 1000\pi t \Rightarrow V_A = 4$ V, $v_a = 5\sin 2000\pi t + 3\cos 1000\pi t$.

Units and conductance

- Equations assume consistent units, e.g. $5 = 10000I_1 + 0.6$ (V, A, Ω).
- Shortcut: **V, mA, kΩ** is also consistent (⇒ power in mW).
- Conductance is the reciprocal of the matching resistor: $G_x = 1/R_x$, $g_\pi = 1/r_\pi$.

Dependent (controlled) sources

Diamond symbol: arrow inside = current source, ± inside = voltage source.

Type	Output	Gain unit	Used for
VCCS	$i = g_m v_1$	S (A/V)	transconductance, FET/BJT small-signal
CCCS	$i = \beta i_1$	A/A	BJT current gain
VCVS	$v = Av_1$	V/V	voltage amplifier, op amp
CCVS	$v = Ki_1$	Ω	transresistance (rarely used)

Handy formulas

- dc value = average: $V_{DC} = \frac{1}{T}\int_0^T v\,dt$. Sine: $V_{pp} = 2V_p$, $V_{rms} = V_p/\sqrt{2}$.
- Phasor: $V_m\cos(\omega t + \phi) \leftrightarrow \mathbf{V} = V_m\angle\phi$. $\sin x = \cos(x - 90^\circ)$.
- $\mathbf{V} = a + jb = |\mathbf{V}|\angle\theta$: $|\mathbf{V}| = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$ (check the quadrant).

1.4 Problem-Solving Approach

[J p.13]

1. **Problem**: restate it clearly
2. **Knowns**: list the given data
3. **Unknowns**: define what to find
4. **Assumptions**: list them (e.g. active region, ideal op amp)
5. **Approach**: choose a method
6. **Analysis**: *draw the circuit and label every V and I*
7. **Check**: is the maths right, are all unknowns found, do the **assumptions still hold**?
8. **Evaluate**: is it realistic? If not, repeat steps 4–7
9. **Simulate** (SPICE) and compare with the hand result

Reasonable numbers (sanity check)

- DC voltages lie between the supply rails, e.g. $V_{EE} \le V \le V_{CC}$. [J p.14]
- AC peak-to-peak ≤ $V_{CC} - V_{EE}$ (e.g. +8/−5 V ⇒ ≤ 13 V_{pp}).
- Currents range from μA to about 100 mA. Max current through R: $I_{max} = V_{supply}/R$.
- Typical values: supplies 10–20 V; R from a few Ω to GΩ.
- If an answer breaks any of these rules, recheck the assumptions (e.g. wrong region).

Handy formulas

- $V = IR$, $P = VI = I^2R = V^2/R$. V/kΩ = mA, mA·kΩ = V, V·mA = mW.
- Prefixes: p 10^{-12}, n 10^{-9}, μ 10^{-6}, m 10^{-3}, k 10^{3}, M 10^{6}, G 10^{9}.

1.5 Important Concepts from Circuit Theory

[J p.15]

Toolkit: KVL, KCL, nodal / mesh analysis, Thévenin / Norton, voltage and current division.

1.5.1 Voltage and current division

Voltage divider (series):

$$v_1 = v_i\frac{R_1}{R_1 + R_2}$$

Only if the same current flows through R_1 and R_2 (nothing else is connected at the middle node).

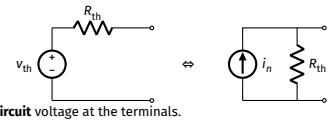
Current divider (parallel):

$$i_1 = i_i\frac{R_2}{R_1 + R_2}$$

Uses the **other** resistor on top. **Only if** the same voltage is across both.

- Parallel: $R_1 \| R_2 = \frac{R_1R_2}{R_1 + R_2}$ $R\|R = R/2$ $R\|10R \approx 0.91R$ (the smaller one dominates).
- AC: replace R with Z and use phasors: $\mathbf{V_1} = \mathbf{V_i}Z_1/(Z_1 + Z_2)$, $\mathbf{I_1} = \mathbf{I_i}Z_2/(Z_1 + Z_2)$. [J p.16]
- **Pitfall**: a load R_L across R_2 breaks the plain divider. Use $R_2 \to R_2\|R_L$.

1.5.2 Thévenin and Norton equivalents



- v_{th} = **open-circuit** voltage at the terminals.
- i_n = **short-circuit** current at the terminals.
- R_{th}: **zero the independent sources** (V source → short, I source → open). **Keep the dependent sources active.** Apply a test source v_x and find i_x: $R_{th} = v_x/i_x$.
- Linked by $v_{th} = i_nR_{th}$: find any two and you get the third. Use this to check your answer.
- With no dependent sources you can find R_{th} by combining resistors directly.

Ex 1.1 pattern (CCCS βi_1 feeding the output node, R_1 from the input): [J p.17–19]

$$v_{th} = \frac{(\beta + 1)R_S}{(\beta + 1)R_S + R_1}v_i \qquad i_n = \frac{(\beta + 1)v_i}{R_1} \qquad R_{th} = R_S \parallel \frac{R_1}{\beta + 1}$$

Insight: the dependent source makes R_1 look ($\beta + 1$) times smaller from the output (emitter-follower idea).

Handy formulas

- Series: $R = R_1 + R_2$. Parallel: $G = G_1 + G_2$ (conductances add).
- Impedance: $Z_R = R$, $Z_L = j\omega L$, $Z_C = \frac{1}{j\omega C}$. ($\omega \to 0$: C open, L short. $\omega \to \infty$: C short, L open.)
- KCL: $\sum i_{in} = \sum i_{out}$ at a node. KVL: $\sum v = 0$ around a loop.
- Source transformation: $v_{th} = i_nR_{th}$. Max power to the load when $R_L = R_{th}$:

$$P_{max} = \frac{v_{th}^2}{4R_{th}}.$$

1.6 Frequency Spectrum of Electronic Signals

[J p.21]

- **Fourier**: any signal is a sum of sinusoids. The **spectrum** is the amplitude and phase of each one versus frequency.
- **Periodic** signal ⇒ **discrete** spectrum (harmonics only). **Non-periodic** ⇒ **continuous** spectrum.
- Fundamental: $\omega_o = 2\pi/T$ (rad/s), $f_o = 1/T$ (Hz). Harmonics are at $2f_o, 3f_o, \ldots$

Square wave (period T, amplitude V_O, offset V_{DC}):

$$v(t) = V_{DC} + \frac{2V_O}{\pi}\left(\sin\omega_o t + \frac{1}{3}\sin 3\omega_o t + \frac{1}{5}\sin 5\omega_o t + \cdots\right)$$

Only **odd** harmonics, with amplitude ∝ $1/n$. V_{DC} is the average (0 Hz) term. ($2V_O/\pi$ means V_O is peak-to-peak; for ±A the factor becomes $4A/\pi$.)

Common signal frequency ranges

[J Table 1.3, p.22]

Audio	20 Hz–20 kHz	AM radio	540–1600 kHz
Baseband video	0–4.5 MHz	FM radio	88–108 MHz
Cellular	0.8–2.7 GHz	Satellite TV	3.7–4.2 GHz

Bandwidth and −3 dB (preview of Bode plots)

[J p.20–21]

- f_L, f_H = lower and upper **cutoff** frequencies, where output **power halves** (−3 dB). There, voltage gain = $1/\sqrt{2} \approx 0.707\times$ the mid-band gain.
- Bandwidth $BW = f_H - f_L$ (e.g. audio amplifier: 20 Hz–20 kHz).

Handy formulas

- $\omega = 2\pi f$ (rad/s ↔ Hz), $f = 1/T$, nth harmonic: $f_n = nf_o$.
- Read f from a signal: $\sin(2000\pi t) \Rightarrow \omega = 2000\pi \Rightarrow f = \omega/2\pi = 1$ kHz.
- dB: $20\log_{10}|A_v|$ (voltage/current), $10\log_{10}(P_2/P_1)$ (power).
- −3 dB = ×0.707 +6 dB = ×2 20 dB = ×10 40 dB = ×100 0 dB = ×1.

1.7 Amplifiers

[J p.22]

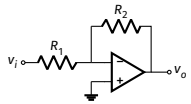
Linear amplifier: same frequency out; only amplitude and phase change. Use **superposition** on each frequency component.
$v_i = V_i\sin(\omega t + \phi) \;\to\; v_o = V_o\sin(\omega t + \phi + \theta)$

$$A_v = \frac{\mathbf{V_o}}{\mathbf{V_i}} = \frac{V_o\angle(\phi + \theta)}{V_i\angle\phi} = \frac{V_o}{V_i}\angle\theta$$

- $|A_v|$ = amplitude ratio, θ = phase shift. Both usually depend on frequency.
- Inverting: $A_v < 0$, i.e. a 180° phase shift (e.g. $A_v = -5$: ×5 and flipped).
- ωt is in **rad**, but books often give ϕ in **degrees**. Convert before calculating.

Ex: $v_i = 0.001\sin(2000\pi t)$, $v_o = -5\cos(2000\pi t + 25^\circ)$ [J p.23]
$-\cos x = \sin(x - 90^\circ) \Rightarrow \mathbf{V_o} = 5\angle -65^\circ$, $\mathbf{V_i} = 0.001\angle 0^\circ$, $A_v = 5000\angle -65^\circ$.

1.7.1 Ideal op amp



$A \to \infty$, $R_{in} \to \infty$ give two rules:
① $v_- = v_+$ ② $i_- = i_+ = 0$

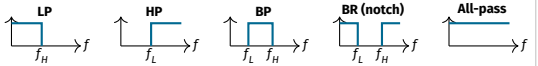
Inverting:

$$A_v = -\frac{R_2}{R_1}$$

$R_{in} = R_1$ [J p.24]

- **Virtual ground**: v_- = 0 because v_+ is grounded, but it is **not** wired to ground. Never short it during analysis. [J p.25]
- Method: KCL at v_- with $i_- = 0 \Rightarrow \frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_2}$.
- Non-inverting (extra): $A_v = 1 + R_2/R_1$ (always ≥ 1, no phase flip). Follower: $A_v = 1$.
- Ex: $R_2 = 100$ kΩ, $A_v = -5 \Rightarrow R_1 = 20$ kΩ.

1.7.2 Amplifier frequency response (ideal filters)



Low-pass	passes $f < f_H$
High-pass	passes $f > f_L$
Band-pass	passes $f_L < f < f_H$, $BW = f_H - f_L$
Band-reject	blocks $f_L < f < f_H$
All-pass	gain is flat at all f; only changes the **phase**

An amplifier designed to pass only certain frequencies is a **filter**. [J p.26]
Method: split v_i into its components, find $f = \omega/2\pi$ for each, keep (times A) the ones in the passband, drop the rest.
Ex: BP filter, $f_L = 1.5$k, $f_H = 2.5$k, $A = 10$.
$v_i = 0.5\sin(2000\pi t) + \sin(4000\pi t) + 1.5\sin(6000\pi t)$ has components at 1, 2, 3 kHz. Only 2 kHz passes, so $v_o = 10\sin(4000\pi t)$.

Handy formulas

- Phasor maths: multiply magnitudes and add angles; divide magnitudes and subtract angles.
- $\cos x = \sin(x + 90^\circ)$, $-\sin x = \sin(x \pm 180^\circ)$, $-\cos x = \sin(x - 90^\circ)$.
- deg → rad: ×π/180. $f = \omega/2\pi$, $T = 1/f$.
- Gain in dB: $A_{dB} = 20\log_{10}|A_v|$. Cascaded stages: gains multiply, dB values add.
- Inverting $-R_2/R_1$ Non-inverting $1 + R_2/R_1$ Follower 1.

1.K Key Takeaways & Watch Outs

[J Summary, p.34]

- **Read subscripts carefully:** v_{BE} total, V_{BE} dc, v_{be} signal, $\mathbf{V_{be}}$ phasor.
- **ω vs f:** the most common slip is the missing 2π. In $\sin(\omega t)$, ω is in rad/s, not Hz.
- **Degrees vs radians:** convert ϕ before putting it into $\sin(\omega t + \phi)$ on a calculator.
- **Dividers** need the same current (voltage divider) or the same voltage (current divider). A load breaks them.
- **Thévenin/Norton:** zero *only* the independent sources and keep the dependent ones. Check with $v_{th} = i_nR_{th}$.
- **Linear amplifiers** create no new frequencies: treat each component separately (superposition).
- **Ideal op amp:** $v_- = v_+$, $i_\pm = 0$ (needs negative feedback). A virtual ground is not a real ground.
- **Filters** are named by their passband. The edges f_L, f_H are the −3 dB (half-power) points.
- **Sanity check** every answer: within the supply rails, currents between μA and about 100 mA, and do the assumptions hold?
- **Answer to 3 significant figures**: component tolerances (1–50%) make more digits meaningless. [J 1.9]

1.F Chapter 1 Formula Sheet

Notation	$v_{BE} = V_{BE} + v_{be}$ $\mathbf{V} = V_m\angle\phi$ $G = 1/R$
V divider	$v_1 = v_i\frac{R_1}{R_1 + R_2}$ $\mathbf{V_1} = \mathbf{V_i}\frac{Z_1}{Z_1 + Z_2}$
I divider	$i_1 = i_i\frac{R_2}{R_1 + R_2}$ $\mathbf{I_1} = \mathbf{I_i}\frac{Z_2}{Z_1 + Z_2}$
Parallel	$R_1\|R_2 = \frac{R_1R_2}{R_1 + R_2}$ $G = G_1 + G_2$
Thévenin	$v_{th} = v_{oc}$ $i_n = i_{sc}$ $R_{th} = \frac{v_x}{i_x} = \frac{v_{th}}{i_n}$
Impedance	$Z_R = R$ $Z_L = j\omega L$ $Z_C = \frac{1}{j\omega C}$
Power	$P = VI = I^2R = V^2/R$ $P_{max} = \frac{v_{th}^2}{4R_{th}}$ at $R_L = R_{th}$
Frequency	$\omega = 2\pi f$ $f = 1/T$ $f_n = nf_o$
Square wave	$V_{DC} + \frac{2V_O}{\pi}\left(\sin\omega_o t + \frac{1}{3}\sin 3\omega_o t + \cdots\right)$
Sine	$V_{rms} = V_p/\sqrt{2}$ $V_{pp} = 2V_p$
Gain	$A_v = \frac{\mathbf{V_o}}{\mathbf{V_i}} = \frac{V_o}{V_i}\angle\theta$ $A_{dB} = 20\log_{10}\|A_v\|$
Op amp	Inv. $-\frac{R_2}{R_1}$ Non-inv. $1 + \frac{R_2}{R_1}$ $R_{in,inv} = R_1$
Bandwidth	$BW = f_H - f_L$ at $f_{L,H}$: gain × $\frac{1}{\sqrt{2}}$, power × $\frac{1}{2}$

1.C Constants & Key Numbers

$1/\sqrt{2}$	0.707 (−3 dB)	$\sqrt{2}$	1.414
2π	6.283	$1/2\pi$	0.159
$2/\pi$	0.637	$4/\pi$	1.273
$\log_{10}2$	0.301 (×2 = 6 dB)	$\pi/180$	0.01745 rad/°
Audio	20 Hz–20 kHz	Supplies	10–20 V typ.
Currents	μA – 100 mA	Sig. figs	3

Physical constants (q, k, V_T, …) will be added with the chapters that use them.

2 Solid-State Electronics

2.1 Solid-State Electronic Materials

[J p.43]

Classified by **resistivity** ρ (Ω·cm). Conductivity $\sigma = 1/\rho$.

Material	ρ (Ω·cm)	e.g.
Insulator	> 10^5	diamond 10^{16}
Semiconductor	10^{-3} – 10^5	Si; ρ is **set by doping**
Conductor	< 10^{-3}	copper 3 × 10^{-6}

- **Elemental**: one element from column IV (Si, Ge, C). **Compound**: III–V (GaAs, InP, GaN) or II–VI (CdSe).
- **Si dominates**: larger bandgap than Ge (works hotter) and a stable insulating oxide (SiO_2) for ICs.
- GaAs, InP: optoelectronics (LEDs, lasers, photodetectors). SiGe: high-frequency BJTs.

Bandgap E_G (eV) [J T2.3]

Si	**1.12**	GaAs	1.42
Ge	0.66	InP	1.35
SiC	3.26	GaN	3.49
C	5.47		

Periodic table [J T2.2]

III	IV	V
B	C	N
Al	Si	P
Ga	Ge	As
In	Sn	Sb

Preview (2.6): III = **acceptors** (p-type), V = **donors** (n-type).

Handy formulas

- $R = \rho\frac{L}{A}$ (ρ in Ω·cm ⇒ L in cm, A in cm²). $\sigma = 1/\rho$ ($\Omega^{-1}\text{cm}^{-1}$).
- 1 cm = 10^4 μm = 10^7 nm. 1 μm = 10^{-4} cm.
- 1 eV = 1.602 × 10^{-19} J. Larger E_G ⇒ fewer carriers ⇒ better insulator / higher max temperature.

2.2 Covalent Bond Model

[J p.44]

- Forms: **amorphous** (disordered), **polycrystalline** (many small crystals), **single-crystal** (used for devices).
- Si: 4 valence electrons, covalent bonds to 4 neighbours (diamond lattice, $a = 0.543$ nm).
- **0 K**: every bond is filled, no free electrons, so Si is an **insulator**.
- $T \uparrow$: thermal energy breaks bonds, each one creating an **electron–hole pair**.
- **Hole**: the vacancy left behind, charge $+q$. It moves as neighbouring electrons hop into it.
- **Bandgap** E_G: the minimum energy needed to break a bond (free an electron).

Intrinsic carrier density [J eq. 2.1, p.45]

$$n_i^2 = BT^3 \exp\left(-\frac{E_G}{kT}\right) \text{ cm}^{-6}$$

T in K, $k = 8.62 \times 10^{-5}$ eV/K, E_G in eV.

	B ($K^{-3}cm^{-6}$)	E_G (eV)	n_i at 300 K (cm^{-3})
Si	1.08×10^{31}	1.12	$6.73 \times 10^9 \approx \mathbf{10^{10}}$
Ge	2.31×10^{30}	0.66	2.27×10^{13}
GaAs	1.27×10^{29}	1.42	–

Ex 2.1 (Si, 300 K):
$n_i^2 = 1.08 \times 10^{31}(300)^3 e^{-1.12/(8.62\times10^{-5}\cdot300)} = 4.52 \times 10^{19} \Rightarrow n_i = 6.73 \times 10^9/\text{cm}^3$. [J p.46]

Intrinsic (pure) material: $n = p = n_i$. In **thermal equilibrium**: [J eq. 2.2, p.47]

$$pn = n_i^2$$

- Holds only with **no external stimulus** (no applied V or I, no light). It also holds for doped material (used later).
- n_i is **very** sensitive to T: for Si it roughly doubles every ~9 K near 300 K.
- Si has 5×10^{22} atoms/cm^3, so only about 1 bond in 10^{13} is broken at room temperature.
- Use $n_i = 10^{10}/\text{cm}^3$ for Si at room temperature unless told otherwise.

Handy formulas

- $T(\text{K}) = T(^\circ\text{C}) + 273$. kT at 300 K = 0.0259 eV. $V_T = kT/q \approx 25.8$ mV (300 K, book value).
- $q = 1.602 \times 10^{-19}$ C. $k = 8.62 \times 10^{-5}$ eV/K $= 1.38 \times 10^{-23}$ J/K.
- Calculator: find n_i^2 first, then take $\sqrt{\ }$. Keep E_G and kT both in eV.
- Ratio between two temperatures: $\frac{n_i^2(T_2)}{n_i^2(T_1)} = \left(\frac{T_2}{T_1}\right)^3 \exp\left[\frac{E_G}{k}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$.

2.3 Drift Currents and Mobility

[J p.47]

2.3.1 Drift current

Drift is carrier motion caused by an electric field **E**. Drift current density:

$$\mathbf{j} = Q\mathbf{v} \quad (\text{C/cm}^3)(\text{cm/s}) = \text{A/cm}^2$$

$\mathbf{j}$ = current density (A/cm^2, per **area**), Q = charge density (C/cm^3, per **volume**), $\mathbf{v}$ = carrier velocity. Current: $I = jA$.

2.3.2 Mobility

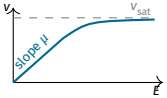
At low fields, velocity is proportional to field. The constant is the **mobility** μ (cm^2/V·s): [J eq. 2.4]

$$\mathbf{v_n} = -\mu_n \mathbf{E} \qquad \mathbf{v_p} = +\mu_p \mathbf{E}$$

- Electrons move **against E** (hence the minus sign); holes move **with E**.
- Intrinsic Si: $\mu_n = \mathbf{1420}$, $\mu_p = \mathbf{470}$ cm^2/V·s. $\mu_n \approx 3\mu_p$: holes are stuck moving through the bond structure, while electrons move freely.
- *Extra:* this is why n-type devices (NMOS, npn) are faster than p-type devices of the same size.

Ex: hole at $E = 10$ V/cm: $v_p = 470 \times 10 = 4.70 \times 10^3$ cm/s. Electron at 1000 V/cm:
$v_n = -1.42 \times 10^6$ cm/s. 1 V across 2 µm: $E = V/L = 1/(2 \times 10^{-4}) = 5 \times 10^3$ V/cm. [J p.48]

2.3.3 Velocity saturation



- $v = \mu E$ only holds for $E \le 5000$ V/cm (0.5 V/µm).
- For $E \ge 3 \times 10^4$ V/cm: $v \to v_{sat} \approx \mathbf{10^7}$ **cm/s** (Si, electrons and holes).
- v_{sat} sets an upper limit on device **frequency response**.

300 K	μ_n (cm^2/V·s)	μ_p	v_{sat} (cm/s)
Si (intrinsic)	1420	470	10^7
Ge	4300	2100	6×10^6
GaAs	8500	–	2×10^7 (e$^-$)

[J p.48, Fig. 2.5]

Handy formulas

- Uniform field: $E = V/L$. 1 V/µm $= 10^4$ V/cm. Keep everything in **cm**.
- $I = jA$. $Q_n = -qn$, $Q_p = +qp$ (C/cm^3). Check units: μE = (cm^2/V·s)(V/cm) = cm/s.
- Transit time across length L: $t = L/v$ (use v_{sat} at high field).
- Preview (2.4): $j_n = qn\mu_n E$, $j_p = qp\mu_p E$, $\sigma = q(n\mu_n + p\mu_p)$, $\rho = 1/\sigma$.

2.4 Resistivity of Intrinsic Silicon

[J p.49]

$$j_n = qn\mu_n E \quad j_p = qp\mu_p E \quad (\text{A/cm}^2)$$

$$j_T = j_n + j_p = q(n\mu_n + p\mu_p)E = \sigma E$$

$$\sigma = q(n\mu_n + p\mu_p) \quad (\Omega \cdot \text{cm})^{-1}$$

$$\rho = \frac{1}{\sigma} = \frac{E}{j_T} \quad (\Omega\text{cm})$$

- Electrons and holes move in opposite directions but carry opposite charge, so their drift currents **add**.
- **Ex 2.2** (intrinsic Si, $n = p = 10^{10}$): $\sigma = 1.6 \times 10^{-19} \cdot 10^{10}(1420 + 470) = 3.02 \times 10^{-6}$, so $\rho = 3.31 \times 10^5$ Ω·cm, a (poor) **insulator**. At 400 K, $\rho = 1420$ Ω·cm, a semiconductor. [J p.50]

Handy formulas

- $R = \rho L/A = L/(\sigma A)$. $V = EL$, $I = jA$.
- Intrinsic: $\sigma_i = qn_i(\mu_n + \mu_p)$. Since n_i rises sharply with T, ρ_i falls sharply with T.

2.5 Impurities in Semiconductors

[J p.50]

Doping: adding tiny, controlled amounts of impurity sets ρ over a huge range and makes the material n- or p-type.

	Donor (2.5.1)	**Acceptor** (2.5.2)
Column	V: **P, As, Sb** (5 valence e$^-$)	III: **B** (3 valence e$^-$)
Effect	extra e$^-$, easily freed (about all ionized at 300 K)	missing bond, which becomes a mobile **hole**
Fixed ion	$+q$ (immobile)	$-q$ (immobile)
Gives	**n-type** (N_D)	**p-type** (N_A)

Memory aid: Donor **D**onates e$^-$ (column V = fiVe). **A**cceptor **A**ccepts e$^-$ (column III), which leaves a hole.

Handy formulas

- Carriers (e$^-$, holes) are **mobile**. Ionized impurities are **fixed** charge in the lattice.
- N_D, N_A in atoms/cm^3. Si has 5×10^{22} atoms/cm^3, so even 10^{18} doping is only about 0.002%.

2.6 Electron and Hole Concentrations in Doped Semiconductors

[J p.51]

n-type: $n > p$. p-type: $p > n$. The larger population is the **majority** carrier; the smaller is the **minority** carrier.

Charge neutrality: [J 2.10]

$$q(N_D + p - N_A - n) = 0$$

Still true for doped material: [J 2.2]

$$pn = n_i^2$$

Exact (n-type, $N_D > N_A$): [J 2.11] $n = \frac{(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2}$, $p = \frac{n_i^2}{n}$. (p-type: swap $N_D \leftrightarrow N_A$ and $n \leftrightarrow p$.) [J 2.12]

Practical (when $|N_D - N_A| \gg 2n_i$, almost always): [J p.53]

$$\text{n-type: } n \cong N_D - N_A, \quad p = \frac{n_i^2}{N_D - N_A}$$

$$\text{p-type: } p \cong N_A - N_D, \quad n = \frac{n_i^2}{N_A - N_D}$$

- Typical doping: $10^{14} \le |N_A - N_D| \le 10^{21}$ /cm^3.
- **Majority** density is set by the doping, so it is **independent of** T. **Minority** density is $\propto n_i^2$, so it is **very** T**-dependent**.
- Why minority carriers matter: diodes and BJTs depend on them (FETs are majority-carrier devices). [J p.54]
- Use the **exact** formula when $|N_D - N_A|$ is close to n_i (very light doping, high T, or near-equal $N_D \approx N_A$).

Ex 2.3: B $= 10^{16}$ (N_A), P $= 2 \times 10^{15}$ (N_D). $N_A > N_D$, so p-type: $p = 8 \times 10^{15}$, $n = 10^{20}/8 \times 10^{15} = 1.25 \times 10^4$ /cm^3. Check: $pn = 10^{20}$. [J p.53]
Ex: Sb $= 2 \times 10^{16}$ is a donor: $n = 2 \times 10^{16}$, $p = 5 \times 10^3$, n-type.

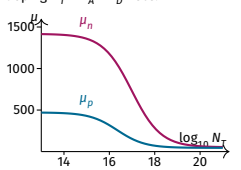
Handy formulas

- Steps: ① identify each impurity as donor (V) or acceptor (III) ② compare N_D vs N_A to find the type ③ majority = $|N_D - N_A|$ ④ minority = n_i^2/majority ⑤ check $pn = n_i^2$.
- Si at 300 K: $n_i^2 = 10^{20}$ cm^{-6}. Minority at another T: recompute $n_i(T)$ first (2.2).

2.7 Mobility and Resistivity in Doped Semiconductors

[J p.54]

Impurities **scatter** carriers (lattice distortion and ionized charge), so μ **falls** as the **total** doping $N_T = N_A + N_D$ rises.



N_T (cm^{-3})	μ_n	μ_p
$\le 10^{14}$	1405	462
10^{15}	1359	430
10^{16}	1177	318
10^{17}	727	153
10^{18}	284	71
10^{19}	108	50
10^{20}	64	46

Si, 300 K, from the Fig. 2.8 equations.

Fig. 2.8 equations (Si, 300 K): [J p.55]

$$\mu_n = 52.2 + \frac{1365}{1 + \left(\frac{N_T}{9.68\times10^{16}}\right)^{0.68}} \qquad \mu_p = 44.9 + \frac{426}{1 + \left(\frac{N_T}{2.23\times10^{16}}\right)^{0.72}}$$

(The book labels both as μ_n; the second is μ_p.)

Majority carriers dominate the conductivity: [J 2.13]

$$\text{n: } \sigma \cong q\mu_n(N_D - N_A)$$

$$\text{p: } \sigma \cong q\mu_p(N_A - N_D)$$

- **Trap:** μ uses the **sum** $N_T = N_A + N_D$, but the carrier density uses the **difference** $|N_D - N_A|$.
- **Ex 2.4**: $N_D = 4 \times 10^{15}$: $n = 4 \times 10^{15}$, $p = 2.5 \times 10^4$, $\mu_n = 1280$, $\mu_p = 375$, so $\sigma = 0.817$ and $\rho = \mathbf{1.22\ \Omega\cdot cm}$ (semiconductor). That is 5 orders of magnitude below intrinsic Si. [J p.56]
- **Ex 2.5** (given ρ, find N_D), **iterate**: guess N_D, get $\mu_n(N_D)$, compare $\mu_n N_D$ with $1/(q\rho)$, repeat. Ex: $\rho = 0.025$ gives $N_D = 7.92 \times 10^{17}$. [J p.57]
- Si becomes a **conductor** ($\rho \le 10^{-3}$) at $N_D \ge 9.68 \times 10^{19}$ (with $\mu_n = 64.5$).
- Ex: B $= 4 \times 10^{18}$: p-type, $n = 25$, $\mu_p = 54.8$, $\rho = 0.0285$ Ω·cm. [J p.58]

Handy formulas

- Full recipe for ρ: type, then n and p (2.6), then μ_n, μ_p at N_T (2.7), then $\sigma = q(n\mu_n + p\mu_p)$, then $\rho = 1/\sigma$.
- The minority term is negligible: e.g. $\mu_p p = 375 \times 2.5 \times 10^4 \ll \mu_n n = 1280 \times 4 \times 10^{15}$.
- Doping both raises the carrier density (a huge effect) and lowers μ (a smaller effect), so ρ still falls overall.

2.8 Diffusion Currents

[J p.58]

Diffusion: carriers spread from **high** to **low** concentration (like smoke in a room). It needs a **gradient**, not a field.

$$j_p^{\text{diff}} = -qD_p \frac{\partial p}{\partial x} \qquad j_n^{\text{diff}} = +qD_n \frac{\partial n}{\partial x} \quad (\text{A/cm}^2)$$

[J 2.14]

- **Signs:** with a positive gradient, both carriers diffuse in $-x$. Holes ($+q$) give current in $-x$, hence the minus sign. Electrons ($-q$) give current in $+x$, hence the plus sign.
- D_n, D_p = **diffusivity** (diffusion coefficient), in cm^2/s. Si at 300 K: $D_n \approx 2$–35, $D_p \approx 1$–15.

Einstein relationship: [J 2.15, p.59]

$$\frac{D_n}{\mu_n} = \frac{kT}{q} = \frac{D_p}{\mu_p} = V_T$$

$$V_T = kT/q = 0.0258 \text{ V at 300 K}$$

V_T: 50 K 4.3 mV, 77 K 6.63 mV, 300 K 25.8 mV, 400 K 34.5 mV. Max D (lightly doped Si): $D_n = 36.6$, $D_p = 12.1$ cm^2/s.

Ex: electron gradient 10^{16}/(cm^3·µm) $= 10^{20}$/cm^4, $D_n = 20$:
$j_n = 1.6 \times 10^{-19} \cdot 20 \cdot 10^{20} = +320$ A/cm^2. Holes, $+10^{20}$/cm^4, $D_p = 4$: $j_p = -64$ A/cm^2.

Handy formulas

- **Gradient units:** per µm → per cm means ×10^4. A linear profile has $\frac{\partial n}{\partial x} = \frac{\Delta n}{\Delta x}$.
- $D = \mu V_T$, e.g. $\mu_n = 1420 \Rightarrow D_n = 1420 \times 0.0258 = 36.6$ cm^2/s.
- $V_T \propto T$: $V_T(T) = 0.0258 \times \frac{T}{300}$ V.

2.9 Total Current

[J p.59]

Total = drift + diffusion, for each carrier: [J 2.16]

$$j_n^T = q\mu_n nE + qD_n \frac{\partial n}{\partial x} \qquad j_p^T = q\mu_p pE - qD_p \frac{\partial p}{\partial x}$$

With Einstein's relation: [J 2.17]

$$j_n^T = q\mu_n n\left(E + \frac{V_T}{n}\frac{\partial n}{\partial x}\right) \qquad j_p^T = q\mu_p p\left(E - \frac{V_T}{p}\frac{\partial p}{\partial x}\right)$$

Combine with **Gauss' law** $\nabla \cdot (\varepsilon E) = Q$ (ε in F/cm) to analyse devices. [J 2.18, p.60]

- *Extra:* in **equilibrium** the net current is 0, so drift and diffusion **cancel** for each carrier. This creates the built-in field of a pn junction (ch. 3).
- Total current: $I = (j_n^T + j_p^T)A$.

Handy formulas

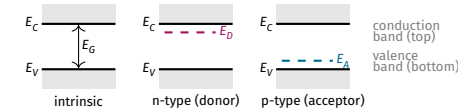
- Drift is driven by E (field) and uses μ. Diffusion is driven by $\partial n/\partial x$ (gradient) and uses D. They are linked by $D = \mu V_T$.
- Uniform doping and no gradient: only drift. No field: only diffusion.

2.10 Energy Band Model

[J p.60]

Conduction band: allowed states for free electrons, bottom edge E_C. **Valence band**: bonding electrons, top edge E_V. No states in between: [J 2.19]

$$E_G = E_C - E_V$$



- **0 K** (2.10.1): valence band full, conduction band empty, so no conduction. $T > 0$: an electron jumps E_G, creating an **electron–hole pair**. This is the band picture of a broken bond. [J p.61]
- **Donor** (2.10.2): level E_D just below E_C. $E_C - E_D \approx 0.045$ eV (P), so almost all donors are ionized at 300 K.
- **Acceptor**: level E_A just above E_V. $E_A - E_V \approx 0.044$ eV (B). Valence electrons fill it, leaving holes.
- **Compensated** (2.10.3, both dopants, $N_D > N_A$): donor electrons drop into the acceptor levels first, so $n = N_D - N_A$.
- The band model and the bond model (2.2) are two views of the same thing.

Handy formulas

- Compare with $kT \approx 0.026$ eV at 300 K: dopant levels (≈ 0.045 eV $\approx 1.7kT$) ionize easily, but $E_G = 1.12$ eV $\approx 43kT$ is hard to cross, which is why n_i is so small.
- In the band picture, electrons fall **down** and holes float **up** (to lower energy).